INCH-POUND

MIL-DTL-24800A
27 March 2026
SUPERSEDING
MIL-DTL-24800
30 September 1996

DETAIL SPECIFICATION

CLEANING COMPOUND, AQUEOUS – OXYGEN SYSTEMS COMPONENTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for an aqueous, inorganic cleaning compound to be used in immersion, ultrasonic, spray, and pumped cleaning of oxygen system piping and components.

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-B-654	-	Brazing Alloys, Silver
QQ-N-281	-	Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections
QQ-N-286	-	Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

COMMERCIAL ITEM DESCRIPTIONS

A-A-50433	-	Grease, Seawater Wash Resistant

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-T-1368	-	Tube and Pipe, Nickel-Copper Alloy, Seamless and Welded
MIL-PRF-5425	-	Plastic Sheet, Acrylic, Heat Resistant
MIL-PRF-6855	-	Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes, General Specification for
MIL-PRF-8625	-	Anodic Coatings for Aluminum and Aluminum Alloy
MIL-PRF-9000	-	Lubricating Oil, Shipboard Internal Combustion Engine, High-Output Diesel

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

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MIL-C-15726	-	Copper-Nickel Alloy, Sheet, Plate, Strip, Bar, Rod and Wire
MIL-T-16420	-	Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706)
MIL-PRF-17331	-	Lubricating Oil, Steam Turbine and Gear, Moderate Service
MIL-PRF-17672	-	Hydraulic Fluid, Petroleum-Based, Inhibited
MIL-DTL-22050	-	Gasket and Packing Material, Rubber, for Use with Polar Fluids, Steam, and Air at Moderately High Temperatures
MIL-DTL-22606	-	Flask and End Plugs, Compressed Gas - Air, Oxygen, and Nitrogen
MIL-S-22698	-	Steel Plate, Shapes and Bars, Weldable Ordinary Strength and Higher Strength: Structural
MIL-T-24107	-	Tube, Copper (Seamless) (Copper Alloy Numbers C10100, C10200, C10300, C10800, C12000, C12200, and C14200)
MIL-PRF-24139	-	Grease, Multipurpose, Water Resistant
MIL-PRF-24508	-	Grease, High Performance, Multipurpose
MIL-PRF-24574	-	Lubricating Fluid for Low and High Pressure Oxidizing Gas Systems
MIL-P-24691	-	Pipe and Tube, Carbon, Alloy and Stainless Steel, Seamless and Welded, General Specification for
MIL-P-24691/3	-	Pipe and Tube, Corrosion-Resistant, Stainless Steel, Seamless or Welded
MIL-PRF-27617	-	Grease, Aircraft and Instrument, Fuel and Oxidizer Resistant

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1330	-	Precision Cleaning and Testing of Shipboard Oxygen, Helium, Helium-Oxygen, Nitrogen, and Hydrogen Systems

(Copies of these publications are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2	-	NAVSEA Hazardous Material Avoidance Process
T9074-BD-GIB-010/0300	-	Base Materials for Critical Applications: Requirements for Low Alloy Steel Plate, Forgings, Castings, Shapes, Bars, and Heads of HY-80/100/130 and HSLA-80/100

(Copies of these publications are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To request an MBPS account, submit a ticket to the ServiceNow Assistance Portal (SNAP), https://snap.navair.navy.mil/mbps (DoD PKI Certificate required). These publications are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

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NATIONAL AERONAUTICS AND SPACE ADMINISTRATION (NASA)

NASA WSTF-TP-629 - Test Protocol and Quality Assurance Program for the Navy Submarine Atmosphere Material Control Program

(Copies of this publication are available from the Lyndon B. Johnson Space Center, White Sands Test Facility, Las Cruces, NM 88004.)

NAVY AND MARINE CORPS FORCE HEALTH PROTECTION COMMAND (NMCFHPC)

NMCPHC TM-6260.01 - Reproductive and Developmental Hazards: A Guide for Occupational Health Professionals

(Copies of this publication are available online at https://www.med.navy.mil.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA 600-4-90-027 - Methods for Measuring the Acute Toxicity of Effluents and Receiving Waters to Freshwater and Marine Organisms

EPA Method-1664 - N-Hexane Extractable Material (HEM; Oil and Grease) and Silica Gel Treated N-Hexane Extractable Material (SGT-HEM; Non-polar Material) by Extraction and Gravimetry

(Copies of these publications are available online at www.epa.gov.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

AIA NAS411-1 - Hazardous Materials Target List

AIA NAS1613 - Seal Element, Packing, Preformed, Ethylene Propylene Rubber

(Copies of these publications are available online at www.aia-aerospace.org.)

ASTM INTERNATIONAL

ASTM B148 - Standard Specification for Aluminum-Bronze Sand Castings

ASTM B150/B150M - Standard Specification for Aluminum Bronze Rod, Bar, and Shapes

ASTM B167 - Standard Specification for Nickel-Chromium-Aluminum Alloys, Nickel-Chromium-Iron Alloys, Nickel-Chromium-Cobalt-Molybdenum Alloy, Nickel-Iron-Chromium-Tungsten Alloy, and Nickel-Chromium-Molybdenum-Copper Alloy Seamless Pipe and Tube

ASTM B171/B171M - Standard Specification for Copper-Alloy Plate and Sheet for Pressure Vessels, Condensers, and Heat Exchangers

ASTM B209/B209M - Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate

ASTM B221 - Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes

ASTM B265 - Standard Specification for Titanium and Titanium Alloy Strip, Sheet, and Plate

ASTM B446 - Standard Specification for Nickel-Chromium-Molybdenum-Niobium Alloy, Nickel-Chromium-Molybdenum-Silicon Alloy, and Nickel-Chromium-Molybdenum-Tungsten Alloy Rod and Bar

ASTM B861 - Standard Specification for Titanium and Titanium Alloy Seamless Pipe

ASTM B862 - Standard Specification for Titanium and Titanium Alloy Welded Pipe

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ASTM D471	-	Standard Test Method for Rubber Property - Effects of Liquids
ASTM D501	-	Standard Test Methods of Sampling and Chemical Analysis of Alkaline Detergents
ASTM D512	-	Standard Test Methods for Chloride Ion in Water
ASTM D543	-	Standard Practices for Evaluating the Resistance of Plastics to Chemical Reagents
ASTM D1193	-	Standard Specification for Reagent Water
ASTM D1248	-	Standard Specification for Polyethylene Plastics Extrusion Materials for Wire and Cable
ASTM D1292	-	Standard Test Method for Odor in Water
ASTM D1331	-	Standard Test Methods for Surface and Interfacial Tension of Solutions of Paints, Solvents, Solutions of Surface-Active Agents, and related Materials
ASTM D1784	-	Standard Specification System and Basis for Specification for Rigid Poly (Vinyl Chloride) (PVC) Compounds and Chlorinated Poly (Vinyl Chloride) (CPVC) Compounds
ASTM D2116	-	Standard Specification for FEP Resin Molding and Extrusion Materials
ASTM D2287	-	Standard Classification System and Basis for Specification for Nonrigid Vinyl Chloride Polymer and Copolymer Molding and Extrusion Compounds
ASTM D3935	-	Standard Classification System and Basis for Specification for Polycarbonate (PC) Unfilled and Reinforced Material
ASTM D4052	-	Standard Test Method for Density, Relative Density, and API Gravity of Liquids by Digital Density Meter
ASTM D4066	-	Standard Classification System for Nylon Injection and Extrusion Materials (PA)
ASTM D4101	-	Standard Classification System and Basis Specification for Polypropylene Injection and Extrusion Materials
ASTM D4894	-	Standard Specification for Polytetrafluoroethylene (PTFE) Granular Molding and Ram Extrusion Materials
ASTM D4895	-	Standard Specification for Polytetrafluoroethylene (PTFE) Resin Produced from Dispersion
ASTM D6778	-	Standard Classification System and Basis for Specification for Polyoxymethylene Molding and Extrusion Materials (POM)
ASTM E70	-	Standard Test Method for pH of Aqueous Solutions with the Glass Electrode
ASTM F483	-	Standard Practice for Total Immersion Corrosion Test for Aircraft Maintenance Chemicals
ASTM F1104	-	Standard Test Method for Preparing Aircraft Cleaning Compounds, Liquid Type, Water Base, for Storage Stability Testing
ASTM G72/G72M	-	Standard Test Method for Autogenous Ignition Temperature of Liquids and Solids in a High-Pressure Oxygen-Enriched Environment

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ASTM G86	-	Determining Ignition Sensitivity of Materials to Mechanical Impact in Ambient Liquid Oxygen and Pressurized Liquid and Gaseous Oxygen Environments: A Modern Day Evaluation of Input Energy and Calibration Variations Between Ambient and Pressurized Liquid and Gaseous Oxygen Environments

(Copies of these publications are available online at www.astm.org.)

SAE INTERNATIONAL

SAE-AMS-M-7866	-	Molybdenum Disulfide, Technical, Lubrication Grade
SAE-AMS-P-5510	-	O-Ring, Preformed, Straight Thread Tube Fitting Boss, Type I Hydraulic (-65 ° to 160 °F)
SAE-AMS-P-83461	-	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Improved Performance at 275 °F (135 °C)
SAE-AMS-QQ-A-250	-	Aluminum and Aluminum Alloy, Plate and Sheet General Specification for
SAE-AMS-QQ-S-763	-	Steel, Corrosion Resistant, Bars, Wire, Shapes, and Forgings
SAE-AMS-R-83285	-	Rubber, Ethylene-Propylene, General Purpose
SAE-AMS-WW-T-700	-	Tube, Aluminum and Aluminum Alloy, Drawn, Seamless, General Specification for
SAE-AMS3216	-	Elastomer: Fluorocarbon (FKM) Rubber High-Temperature - Fluid Resistant Low Compression Set/70 to 80 Type A Hardness for Products in Fuel Systems and Specific Engine Oil Systems
SAE-AMS3218	-	Elastomer: Fluorocarbon (FKM) Rubber High-Temperature-Fluid Resistant Low Compression Set/85 to 95 Type A Hardness for Seals in Fuel Systems and Specific Engine Oil Systems
SAE-AMS7259	-	Rubber: Fluorocarbon (FKM) High Temperature/Fluid Resistant Low Compression Set/85 to 95 Hardness for Seals in Fuel Systems and Specific Engine Oil Systems
SAE-AMS-7276	-	Rubber: Fluorocarbon (FKM) High Temperature - Fluid Resistant Low Compression Set/70 to 80 Type A Hardness for Seals in Fuel Systems and Specific Engine Oil Systems
SAE-AMS8660	-	Silicone Compound, NATO Code Number S-736

(Copies of these publications are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Composition. The cleaning compound shall be an aqueous, inorganic solution meeting the performance characteristics of 3.4. The cleaning composition shall not intentionally contain carbon-hydrogen bonded material or material that is not completely soluble. Examples of prohibited material include polar organic surface active agents, polar solvents, or dispensable fluorocarbon surface active agents. If any of these prohibited materials are present as a result of being a trace impurity in another ingredient(s), the concentration shall conform to the requirements specified in 3.4.3.

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3.3 Navy oxygen cleaner (NOC).

3.3.1 NOC composition. The NOC composition shown in table I shall meet the performance characteristics of 3.4 when manufactured as specified in 3.3.2 and packaged as specified in 5.1.

TABLE I. Nominal NOC composition.

Component [1/]	Percent by weight
Polysilicate solution	25.0
Water	73.0
Sodium molybdate dihydrate	0.5
Fluoboric acid [2/]	1.5

NOTES:

1/ Components shall be added in the order listed.

2/ Fluoboric acid shall be added to the batch diluted at least 5:1 with water at a rate not greater than 2 gallons per minute. The batch shall be recirculated while the fluoboric acid is being added.

3.3.2 NOC manufacture. Unless otherwise approved (see 6.2), NOC shall be blended in welded stainless steel tanks, piping and components shall be stainless steel, hoses shall be Teflon-lined, and gaskets shall be Viton or Teflon. The cleaning compound shall be filtered through a 0.1-micron membrane filter. Filters shall be constructed of stainless steel, Teflon, polysulfone, or polyethylene, and shall contain no plasticizer, adhesive, or other extractable material. Production of the cleaning compound shall utilize dedicated equipment that shall not be used for any other purpose. A recirculating pump, or other approved method (see 6.2), shall be used to ensure complete blending of the cleaning compound. The minimum blending time when recirculating NOC through a pump shall be determined as follows:

$$\text{Minimum blending time} = (0.693/W)(V)(5)$$

where: W = pump flow rate in gallons per minute
V = batch tank volume in gallons

3.3.3 NOC raw materials. Raw materials used for the NOC composition cleaning compound shall meet the requirements shown in table II when tested as specified in table X.

TABLE II. NOC raw material requirements.

Component	Property	Requirement
Polysilicate	Appearance	Clear and colorless
	Insoluble matter (wt% [1/], max)	0.1
	SiO_2/Na_2O weight ratio	1.80 to 2.00
	Total silica (wt% SiO_2)	24.1 to 27.0
	Total alkalinity (wt% Na_2O)	13.4 to 14.2
	Specific gravity (°BE [2/])	44.1 to 45.1
Water	Reagent grade water	ASTM D1193, type II
	Chlorides (ppm, max)	<2.0
	Total organic carbon (mg/L, max)	1.0

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TABLE II. NOC raw material requirements – Continued.

Component	Property	Requirement
Sodium molybdate dihydrate	Assay (wt%)	99.5 to 103.0
	pH of 5 wt% solution	7 to 10
	Insoluble matter (wt%, max)	0.005
Fluoboric acid	Assay (wt%)	48 to 52
	Boric acid content (wt%, max)	2.0
	Specific gravity at 77 °F (25 °C) (min)	1.365

NOTES:

1/ Percent by weight.

2/ Degrees Baumé.

3.3.4 NOC in-process requirements. In-process verification of the NOC composition cleaning compound shall meet the requirements shown in table III when tested as specified in table XI. Adjustments to meet the requirements of table III shall only be accomplished with raw materials meeting the requirements of 3.3.3.

TABLE III. NOC in-process requirements.

Property	Requirement
Polysilicate anion concentration (wt%)	9.3 to 10.3
Fluoboric acid concentration (wt%)	1.5±0.15
Sodium molybdate dihydrate concentration (wt%)	0.5±0.05
Chlorides (ppm, max)	See 3.4.16
pH	11.90 to 11.99
Total alkalinity (mL NOC, max)	15.0
Insoluble matter (mg/L, max)	See 3.4.17
NVR by filtration (mg/L, max)	See 3.4.13
Surface tension (dynes/cm^2)	50 to 60
Specific gravity (77/68 °F [25/20 °C])	1.090 to 1.1055
Odor	See 3.4.18
Foam	See 3.4.11
Appearance	See 3.4.19

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3.3.5 NOC end-item requirements. End-item verification of the NOC composition cleaning compound shall meet the requirements shown in table IV when tested as specified in table XII.

TABLE IV. NOC end-item requirements.

Property	Requirement
pH	11.90 to 11.99
Total alkalinity (mL NOC, max)	15.0
NVR by filtration (mg/L, max)	See 3.4.13
Odor	See 3.4.18
Foam	See 3.4.11
Appearance	See 3.4.19

3.4 Performance characteristics.

3.4.1 Cleaning efficiency. Cleaning efficiency shall be determined by the ability of the cleaning compound to remove each of the materials (soils) specified in table V through precision ultrasonic, high soil loading, and pipe cleaning as specified below.

TABLE V. Cleaning efficiency.

Soil	Specification	Cleaning efficiency (% removal by weight)		
		Precision ultrasonic cleaning (see 3.4.1.1)	High soil loading cleaning (see 3.4.1.2)	Pipe cleaning (see 3.4.1.3)
Lubricating oil	MIL-PRF-9000 and MIL-PRF-17331	99.0	98.0	99.0
Hydraulic oil	MIL-PRF-5606 [3/] and MIL-PRF-17672	99.0	98.0	99.0
Waterproof grease	A-A-50433	99.0	95.0	99.0
Waterproof grease	MIL-PRF-24139 and MIL-PRF-24508	99.0	75.0	35.0
CTFE [1/] fluorinated oil	Commercial	99.0	98.0	99.0
CTFE fluorinated grease with silica filler	Commercial	99.0	95.0	90.0
CTFE fluorinated grease with halocarbon filler	Commercial	99.0	80.0	45.0
PFPE [2/] fluorinated oil	MIL-PRF-24574	99.0	98.0	99.0
PFPE fluorinated grease	MIL-PRF-27617, type III	99.0	50.0	95.0
Molycote	SAE-AMS-M-7866	99.0	-	99.0
Silicone oil, 30K cp	Commercial	99.0	95.0	99.0

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TABLE V. Cleaning efficiency – Continued.

Soil	Specification	Cleaning efficiency (% removal by weight)		
		Precision ultrasonic cleaning (see 3.4.1.1)	High soil loading cleaning (see 3.4.1.2)	Pipe cleaning (see 3.4.1.3)
Silicone grease with molycote filler	Commercial	99.0	95.0	99.0
Silicone grease	SAE-AMS8660	99.0	45.0	99.0

NOTES:

1/ Chlorotrifluoroethylene.

2/ Perfluoropolyether.

3/ MIL-PRF-5606 is inactive for new design and intended only for guidance. It will eventually be replaced.

3.4.1.1 Precision ultrasonic cleaning. The cleaning compound shall remove not less than the percentage specified in table V of the soil weight deposited on each test coupon when tested as specified in 4.5.1.

3.4.1.2 High soil loading cleaning. The cleaning compound shall remove not less than the percentage by weight specified in table V of the soil deposited on each test coupon when tested as specified in 4.5.1.

3.4.1.3 Pipe cleaning. The cleaning compound shall remove not less than the percentage by weight specified in table V of the soil deposited on each test coupon when tested as specified in 4.5.2.

3.4.2 Flammability. The cleaning compound shall not be ignitable in the presence of a 2,000 pounds per square inch gauge (psig) oxygen atmosphere at a temperature of 800 °F (425 °C) when tested in accordance with ASTM G72/G72M. In addition, the cleaning compound shall not react with liquid oxygen when tested in accordance with ASTM G86.

3.4.3 Toxicity. When evaluated in accordance with 4.5.12, the NOC shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.5.12 and 6.9).

3.4.4 Prohibited materials. The NOC shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or AIA NAS411-1 for all other programs.

3.4.5 Off-gassing limits. When tested in accordance with NASA WSTF-TP-629, the cleaning compound shall evolve none of the following:

a. Detectable halogenated compounds, ammonia, or mercury at 100 °F (38 °C) and at 180 °F (82 °C).

b. Individual hydrocarbon or inorganic compounds greater than 1.0 parts per million (ppm) by volume at 100 °F (38 °C) and at 180 °F (82 °C).

c. Total hydrocarbon (as methane) greater than 5.0 ppm by volume at 100 °F (38 °C).

d. Total hydrocarbon (as methane) greater than 10.0 ppm by volume at 180 °F (82 °C).

3.4.6 Effluent toxicity limit. The cleaning compound shall not exhibit a TLm_{96} aquatic LC_{50} less than 5,000 ppm, when tested for effect on flathead minnows (Pimephales promelas) in accordance with EPA 600-4-90-027.

3.4.7 Medical review. The cleaning compound, the cleaning compound composition, the results of the NASA TP-WSTF-629 at 100 °F (38 °C) and at 180 °F (82 °C), the effluent toxicity results, and the product Safety Data Sheet (SDS) shall be submitted to the Bureau of Medicine and Surgery (BUMED) and Navy and Marine Corps Force Health Protection Command (NMCFHPC) for toxicity review. The cleaning compound shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the preparing activity to the appropriate medical service who will act as advisor to the contacting agency.

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3.4.8 Corrosivity. The cleaning compound shall cause no weight loss in excess of the limits specified in table VI when tested as suggested in ASTM D1280 (for guidance only, see 6.2) or in accordance with ASTM F483, except that immersion shall be for 24 hours at 170±2 °F (77±l °C).

TABLE VI. Total immersion corrosion limits.

Alloy	Specification	Limit ($mg/cm^2/24hrs$)
Aluminum 5052-0	SAE-AMS-WW-T-700	0.70
Aluminum 5456	ASTM B209/B209M, HT116	0.10
Aluminum 6061-T6	ASTM B221	0.04
Aluminum 2024-T3 1/	SAE-AMS-QQ-A-250	0.04
Brazing material BCuP-5	QQ-B-654	0.50
Bronze CDA922	ASTM B584 or ASTM B505	0.10
Copper C12200-H80	MIL-T-24107	0.60
Copper-nickel C715-70/30	MIL-T-16420	0.10
	MIL-C-15726	0.04
Inconel N06600 or similar	ASTM B167, cold-worked	0.10
	ASTM B167, hot-finished	0.04
Inconel N06625 or similar	ASTM B446, cold-rolled	0.04
Naval brass 464	ASTM B171/B171M	0.20
Nickel-aluminum-bronze C95800	ASTM B148	0.50
Nickel-aluminum-bronze C63000-HR50	ASTM B150/B150M	0.04
Nickel-aluminum-bronze C64200-HR50	ASTM B150/B150M	0.60
Nickel-copper	MIL-T-1368	0.50
	QQ-N-281 CL-A	0.60
	QQ-N-286 cold aged	0.70
Stainless steel 304	MIL-P-24691 and MIL-P-24691/3	0.04
	SAE-AMS-QQ-S-763 cold	0.70
Stainless steel 316	SAE-AMS-QQ-S-763 cold	0.08
Steel, ABS grade EH36	MIL-S-22698	0.08
Steel HY-100 (plate)	T9074-BD-GIB-010/0300	0.04
Steel HY-100 (castings)	T9074-BD-GIB-010/0300	0.80
Steel 4130	MIL-DTL-22606	0.04
Titanium, pure	ASTM B861 and ASTM B862	0.04
Titanium 6A1-4V	ASTM B265 GR5	0.04

NOTE:

1/ Sulfuric acid anodized in accordance with MIL-PRF-8625 and sealed by boiling in demineralized water for 15 minutes.

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3.4.9 Rubber compatibility. The cleaning compound shall cause no change in excess of that shown in table VII when tested in accordance with ASTM D471, except that immersion shall be for 24 hours at 170±2 °F (77±1 °C).

TABLE VII. Rubber compatibility limits.

Rubber	Specification [1/]	Durometer hardness (change in units)	Tensile strength (% change)	Ultimate elongation (% change)	Volume (% change)
Buna-N	SAE-AMS-P-5510	6	5	10	1.7
	MIL-DTL-5516 [2/] - class A	1	10	12	1.3
	MIL-DTL-5516 [2/] - class B	2	20	20	1.5
	MIL-P-25732 [2/]	2	20	20	1.5
	SAE-AMS-P-83461	2	10	12	0.4
Buna-S	MIL-S-21923 [2/]	1	12	20	1.2
Polychloroprene	MIL-PRF-6855	3	5	15	3.0
Hypalon-40	MIL-R-81828 [2/]	6	15	10	2.5
Ethylene	MIL-DTL-22050 - 65DH	4	5	5	0.5
Propylene	MIL-DTL-22050 - 80DH	4	5	5	1.5
	AIA NAS1613 - 70DH	6	20	12	1.7
	AIA NAS1613 - 80DH	4	35	12	0.5
	SAE-AMS-R-83285 - 60DH	2	15	5	1.2
	SAE-AMS-R-83285 - 80DH	1	5	5	1.3
KEL-F	Commercial	3	5	45	0.2
Thiokol FA	Commercial	6	10	20	1.2
Adiprene	Commercial	7	55	85	2.0
Viton	SAE-AMS7276 and SAE AMS3216 - 75DH	6	15	25	2.3
	SAE-AMS7259 and SAE SAE-AMS3218 - 90DH	4	10	30	0.7
Silicone	Commercial	1	5	55	0.2

NOTES:

[1/] DH = Durometer hardness.

[2/] Reference listed for historical purposes only. New materials will need to be tested for compatibility in accordance with MIL-STD-1330.

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3.4.10 Plastic compatibility. The cleaning compound shall cause no change in excess of that shown in table VIII when tested in accordance with ASTM D543, except that immersion shall be for 24 hours at 170±2 °F (77±l °C).

TABLE VIII. Plastic compatibility limits.

Plastic	Specification	Tensile strength (% change)	Weight (% change)
Delrin (polyacetal)	ASTM D6778	5.0	0.1
Acrylonitrile butadiene styrene (ABS)	Commercial	5.0	1.2
Epoxy, unfilled	Commercial	25	2.2
Fluorinated ethylene propylene (FEP)	ASTM D2116	5.0	0.1
Surlyn	Commercial	25	1.0
Acrylic	MIL-PRF-5425, finish A	25	1.5
Polyamide (Nylon 6, 6)	ASTM D4066	40	6.0
Polycarbonate (Lexan)	ASTM D3935	5.0	0.3
Polyethylene, HD	ASTM D1248	6.0	0.1
Polypropylene, unfilled	ASTM D4101	7.0	0.1
VESPEL (polyimide), graphite filled	Commercial	5.0	0.7
Polytetrafluoroethylene (PTFE)	ASTM D4894, ASTM D4895	15	0.1
Polyurethane	Commercial	6.0	2.5
Polyvinyl chloride (PVC) rigid	ASTM D1784	5.0	0.5
Polyvinyl chloride (PVC) non-rigid	ASTM D2287	10	1.0
Polyetherimide (injection molded)	Commercial	5	1.0
Polyamide-imide (injection molded)	Commercial	10	1.0

3.4.11 Foaming. The cleaning compound shall exhibit no visible foam when tested as specified in 4.5.3.

3.4.12 Oil and grease. The cleaning compound shall contain no more than 1.0 ppm of oil or grease when tested as specified in table IX.

3.4.13 Non-volatile residue (NVR). The cleaning compound shall contain no more than 1.0 ppm filterable NVR when tested as specified in appendix A.

3.4.14 High temperature stability. The compound shall not precipitate, separate, or exhibit other non-homogeneities after being held at the manufacturer's recommended maximum operating temperature or 175±2 °F (80±l °C), whichever is greater, for 168 hours and allowed to return to standard conditions when tested as specified in 4.5.4.

3.4.15 Low temperature stability. The compound shall not precipitate, separate, or exhibit other non-homogeneities after being held at 37±2 °F (2.5±l °C) for 168 hours and allowed to return to standard conditions when tested as specified in 4.5.5.

3.4.16 Chlorides. The cleaning compound shall have no greater than 2.0 ppm chloride concentration when tested as specified in table IX.

3.4.17 Insoluble matter. The cleaning compound shall contain no greater than 1.0 ppm insoluble matter when tested as specified in 4.5.8.

3.4.18 Odor. The cleaning compound shall have no odor when tested as specified in 4.5.9.

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3.4.19 Appearance. The cleaning compound shall be a clear, water white liquid free of any oil or solids when tested as specified in 4.5.10.

3.4.20 Extended storage stability. After storage for 1 year as specified in 4.5.11, the cleaning compound shall meet the performance requirements of 3.4.11 through 3.4.19 when tested as specified in table IX.

3.4.21 Service tests. Service tests of production samples shall be initiated upon completion of all laboratory tests at Government field activities to demonstrate the ability of the cleaning composition to clean oxygen piping and components using the aqueous oxygen cleaning process in accordance with MIL-STD-1330 or other Government-approved aqueous oxygen cleaning processes specified (see 6.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3 and 6.3).

b. Conformance inspection (see 4.4 and 6.4).

4.2 Inspection conditions. Unless otherwise specified in the test method, all inspections shall be performed in accordance with standard conditions. Standard condition shall be a temperature of 72±4 °F (22±2 °C) and a relative humidity of 50±20 percent.

4.3 First article inspection. First article inspection shall be performed on a representative production sample when a first article sample of sufficient size to complete all laboratory testing is required (see 3.1). This inspection shall include the tests specified in table IX. The first article inspections for an alternative product (compositions other than as specified in 3.3.1) shall consist of all the tests specified in table IX.

TABLE IX. First article inspections.

Property	Requirement	Test method
Flammability	3.4.2	ASTM G72/G72M and ASTM G86
Toxicity:	3.4.3	See 4.5.12
Off-gas toxicity	3.4.5	NASA WSTF-TP-629
Effluent toxicity	3.4.6	EPA 600-4-90-027
Medical review	3.4.7	1/
Cleaning efficiency:	3.4.1	
Precision ultrasonic	3.4.1.1	See 4.5.1
High soil loading	3.4.1.2	See 4.5.1
Pipe cleaning	3.4.1.3	See 4.5.2
Corrosivity	3.4.8	2/ ASTM D1280 3/ or ASTM F483
Rubber compatibility	3.4.9	2/ ASTM D471
Plastic compatibility	3.4.10	2/ ASTM D543
Foaming	3.4.11	See 4.5.3
Oil and grease	3.4.12	4/ ASTM D2579 3/, ASTM D4779 3/, ASTM D3921 3/, ASTM D4281 3/, or EPA Method-1664
NVR	3.4.13	See appendix A 4/
High temperature stability	3.4.14	See 4.5.4
Low temperature stability	3.4.15	See 4.5.5
Chlorides	3.4.16	ASTM D512

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TABLE IX. First article inspections – Continued.

Property	Requirement	Test method
Insoluble matter	3.4.17	See 4.5.8
Odor	3.4.18	See 4.5.9
Appearance	3.4.19	See 4.5.10
Extended storage stability	3.4.20	See 4.5.11
Service tests	3.4.21	MIL-STD-1330 5/

NOTES:

1/ Performed by BUMED and NMCFHPC.

2/ Immersion time shall be for 24 hours at 170±2 °F (77±1 °C).

3/ Intended only for guidance. Use of document(s) or alternative testing procedures requires Technical Authority approval (see 6.2, 6.6.1, and 6.6.2).

4/ Equivalent military or commercial test methods may be used with prior Government approval.

5/ Conducted at approved Government field activities. Additional Government-approved test methods may be specified (see 6.2).

4.3.1 First article service test samples. Production samples shall be provided for service tests. The samples shall be packaged as specified in 5.1. Samples shall be forwarded in accordance with instructions contained in the authorizing letter granting the service test, which will be sent to the manufacturer on satisfactory completion of all laboratory tests. The identification of each drum shall be as follows:

a. Cleaning Compound - Aqueous - Oxygen Systems Components.

b. First article service test samples.

c. Specification MIL-DTL-24800.

d. Manufacturer’s name and product number.

e. Submitted by (name and date) for first article testing in accordance with authorization (reference authorizing letter).

4.4 Conformance inspections. Manufacture of NOC products shall consist of the following conformance inspections:

a. Raw material inspections (see 4.4.2).

b. In-process inspections (see 4.4.3).

c. End-item inspections (see 4.4.4).

4.4.1 NOC lot formation. Unless otherwise specified (see 6.2), a lot shall consist of all the cleaning compound produced by one supplier, at one plant, from the same materials, under essentially the same manufacturing conditions, provided the operation does not exceed 24 hours. When the process is considered a batch operation, each batch shall constitute a lot.

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4.4.2 NOC raw materials inspections. Each lot or batch of raw materials used for the production of NOC compound shall be inspected as specified in table X. Failure of a sample to meet the requirements specified in table II shall be cause for rejection of the entire lot or batch of applicable raw materials.

TABLE X. NOC raw materials inspections.

Component	Property [1/]	Test method
Polysilicate	Appearance	See 4.5.10
	Insoluble matter	ASTM D501 section 56-57
	SiO_2/Na_2O ratio	ASTM D501
	Total silica	ASTM D501 section 51-53
	Total alkalinity	ASTM D501 section 48-50
	Specific gravity	ASTM D4052
Water	Reagent grade water	ASTM D1193
	Chlorides	ASTM D512 method C
	Total organic carbon	ASTM D2579 and ASTM D4779
Sodium molybdate dihydrate	Assay	ACS method
	pH (5% solution)	ASTM E70
	Insoluble matter	ASTM D501 section 88-89
Fluoboric acid	Assay	See 4.5.6
	Boric acid content	See 4.5.6
	Specific gravity	ASTM D4052

NOTE:

1/ Property requirements are in table II.

4.4.3 NOC in-process inspections. Each lot or batch of NOC compound produced shall be inspected as specified in table XI. A sample shall be taken no longer than 168 hours prior to end-item packaging. Failure of the sample to meet the requirements specified in table III shall be cause for rejection of the entire lot or batch.

TABLE XI. NOC in-process inspections.

Property [1/]	Test method
Polysilicate anion concentration	See 4.5.7
Fluoboric acid concentration	ICP-AA [2/]
Sodium molybdate dihydrate concentration	ICP-AA [2/]
Chlorides	ASTM D512
pH	See appendix B
Total alkalinity	See appendix C
Surface tension	ASTM D1331
Specific gravity (77/68 °F [25/20 °C])	ASTM D4052
Insoluble matter	See 4.5.8

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TABLE XI. NOC in-process inspections – Continued.

Property [1/]	Test method
NVR by filtration	See appendix A
Odor	See 4.5.9
Foam	See 4.5.3
Appearance	See 4.5.10

NOTES:

1/ Property requirements are in table III.

2/ Using inductively coupled plasma (ICP) atomic absorption (AA) spectroscopy and assuming that all molybdenum is in the form of sodium molybdate dihydrate and that all boron is in the form of fluoboric acid.

4.4.4 NOC end-item inspections. The first, last, and every fifth end-item 55-gallon container or every tenth end-item 5-gallon container from each lot or batch shall be tested as specified in table XII. Failure of any selected sample to meet the requirements specified in table IV shall be cause for rejection of the entire lot or batch.

TABLE XII. NOC end-item inspections.

Property [1/]	Test method
pH	Appendix B
Total alkalinity	Appendix C
NVR by filtration	Appendix A
Odor	4.5.9
Foam	4.5.3
Appearance	4.5.10

NOTE:

1/ Property requirements are in table IV.

4.5 Methods of inspection.

4.5.1 Cleaning efficiency (precision and high soil loaded).

4.5.1.1 Test apparatus. An ultrasonic cleaner with generator power of no less than 3 watts per square inch of radiated surface, generator frequency of no less than 25 kilohertz (kHz), and capable of maintaining the temperature of its contents at 155±5 °F (68±3 °C) shall be used.

4.5.1.2 Specimen preparation. Three Monel or similar coupons, 1.0 by 4.0 by thickness of 0.125 to 0.250 inch, shall be cleaned by wiping with laboratory grade tissue wet with reagent grade acetone. The coupons shall be thoroughly dried then weighed (W_1). One surface of each of the three coupons shall be coated with a soil (10±1 milligrams [mg] for precision cleaning [see 3.4.1] and 200±5 mg for high soil loading [see 3.4.1.2]) in table V. The coated coupon weight shall be recorded as W_2.

4.5.1.3 Procedure. The three coupons shall be immersed in the operating ultrasonic tank maintained at 155±5 °F (68±3 °C) for no longer than 15 minutes. The coupons shall then be removed and immersed in an unagitated tank of deionized water at 105±5 °F (40±3 °C) for 15±5 minutes, followed by drying in an oven at 220±5 °F (105±3 °C) for 25±5 minutes. The coupons shall be weighed (W_3). This procedure shall be repeated for

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each soil and weight condition. Each cleaning efficiency shall be calculated as follows with the average of the three coupons reported:

$$\text{Percent cleaning efficiency} = [(W_2-W_3)/(W_2-W_1)](100)$$

4.5.2 Pipe cleaning efficiency.

4.5.2.1 Pipe cleaning system. A pipe flow test system shall be constructed that will permit a flow of cleaning compound of 6 feet per second (ft/sec) through the pipe flow cleaning fixture at 160±5 °F (71±3 °C) for 60±5 minutes. The pipe flow test system can be either closed loop (to recirculate the compound) or open loop (no recirculation).

4.5.2.2 Pipe cleaning fixture. The pipe cleaning fixture shall consist of two ¾-inch diameter pipes perpendicular to each other in a pipe tee. Each pipe shall be a minimum of 12 inches in length. Contamination carriers shall be fabricated by welding or brazing 5-inch, ¼-20 threaded stainless steel or brass rod to ¾-inch threaded pipe plugs. The fixture shall be installed into the cleaning system so that flow is perpendicular to the threaded rod when the contamination carrier is inserted into the unused port of the pipe tee fitting.

4.5.2.3 Procedure. The pipe flow contamination carrier shall be cleaned by scrubbing with a nylon brush, wet with reagent grade acetone. The carrier shall be dried and the weight recorded as W_1. Two contamination carriers shall be evenly coated with 200±5 mg of a soil from table V, except for Molycote, which shall be 20±5 mg. The coated carriers shall be weighed (W_2). The contamination carrier shall be inserted into the unused port of the pipe flow fixture and be flushed with 160±5 °F (71±3 °C) cleaning compound flowing at 6 ft/sec for 60±5 minutes. The contamination carrier shall be removed from the fixture and immersed in an unagitated tank of deionized water at 105±5 °F (40±3 °C) for 15±5 minutes, followed by drying in an oven at 220±5 °F (105±3 °C) for 25±5 minutes. The dried carrier shall then be reweighed (W_3). The procedure shall be repeated twice for each soil. Each pipe cleaning efficiency shall be calculated as follows with the average of the two contamination carriers reported:

$$\text{Percent cleaning efficiency} = [(W_2-W_3)/(W_2-W_1)](100)$$

4.5.3 Foaming. A 500-milliliter (mL) quantity of cleaning compound shall be transferred to a clean 1-liter container. The jar shall be capped and vigorously shaken by hand for 15 seconds, then allowed to stand undisturbed for 5 minutes, then visually examined for any evidence of foam.

4.5.4 High temperature stability. A 500-mL quantity of cleaning compound shall be placed in a clean 1-liter Teflon-lined glass or stainless steel container, loosely covered, then heated to the manufacturer's recommended maximum operating temperature or 175±2 °F (80±l °C), whichever is greater, and maintained at that temperature for 168 hours. The cleaning compound shall then be allowed to cool to room temperature while still in the container, then visually examined for evidence of precipitation, separation of other non-homogeneity.

4.5.5 Low temperature stability. A 500-mL quantity of cleaning compound shall be placed in a clean 1-liter Teflon-lined glass or stainless steel container, capped, then placed in a cold chamber at 37±2 °F (2.5±l °C) for 168 hours. The cleaning compound shall then be allowed to warm to room temperature while still in the container, then visually examined for evidence of precipitation, separation of other non-homogeneity.

4.5.6 Fluoboric acid.

4.5.6.1 Reagents.

a. 30 percent calcium chloride solution (adjusted to pH 4.0).

b. Demineralized water (ASTM D1193, type II).

c. 0.1 percent methyl-orange solution.

d. 1 percent phenolphthalein solution.

e. Sodium hydroxide solution (1.0 N).

f. Sorbitol.

g. Mannitol-D powder (commercial edible number 10).

4.5.6.2 Procedure. 1.0±0.5 grams of fluoboric acid (weight A) shall be diluted with 100 mL of demineralized water in a 500-mL reflux flask. 50 mL of 30 percent calcium chloride solution shall be added and refluxed for 1 hour. After cooling to ambient, contents shall be transferred to a 500-mL beaker. Six drops of

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0.1 percent methyl-orange solution and six drops of 1 percent phenolphthalein solution shall be added. 1.0 N sodium hydroxide solution (normality B) shall be titrated until a phenolphthalein red end-point is observed or pH 8.0±0.1 when measured in accordance with ASTM E70. The quantity of titrate (volume C) shall be recorded. 10±1 grams of mannitol-D powder or sorbitol shall be added, stirred, and titrated with 1.0 N sodium hydroxide solution (normality B) to a phenolphthalein red end-point or pH 8.0±0.1 when measured in accordance with ASTM E70. The quantity of titrate (volume D) shall be recorded. The fluoboric acid and boric acid content shall be calculated as follows:

$$\text{Percent fluoboric acid} = [(C)(B)(2.195)]/(A)$$

$$\text{Percent boric acid} = [(D)(B)(6.184)]/[(A)(9.534)]$$

4.5.7 Polysilicate anion concentration. Twenty grams of the NOC cleaning compound (weight = A) shall be transferred to a 250-mL beaker. 80 mL of ASTM D1193, type II demineralized water shall be added. A pH titration with 1.0 N hydrochloric acid solution (normality = B) shall be performed using 0.5-mL increments until 30 mL of hydrochloric acid solution have been added. The pH shall be plotted against the volume of titrant and the volume required to reach the second inflection point (volume = C) shall be determined. The second inflection point should occur between a pH of 4 and 3. The polysilicate anion concentration shall be calculated as follows:

$$\text{Percent polysilicate anion} = [(10.7)(B)(C/A)] - (D)(E) - (F)(G)$$

where: D is the fluoboric acid concentration from 4.4.3.
E is the fluoboric acid assay from 4.4.2.
F is the sodium molybdate concentration from 4.4.3.
G is the sodium molybdate assay from 4.4.2.

4.5.8 Insoluble matter. The test procedure shall be performed as specified in appendix A, with the following exceptions:

a. Step A.5.7 – the oven shall maintain a temperature of 212 to 221 °F (100 to 105 °C).

b. Step A.7.1 – two 500- to 600-mL NOC samples shall be collected.

c. Step A.7.2 – the total sample size shall be 1 liter.

d. Steps A.9.d through A.9.j – testing shall be performed a total of five times.

e. Step A.9.k – the oven temperature shall be 212 to 221 °F (100 to 105 °C), and the time shall be 50 to 60 minutes.

f. Step A.10.b – the insoluble matter shall be as measured in step A.10.a.

4.5.9 Odor. Odor shall be determined in accordance with ASTM D1292, except that no dilution shall be performed and the use of more than one tester is not required.

4.5.10 Appearance. Appearance shall be determined by visually examining 500 mL of cleaning composition for clarity, color, oil, and solids in a clear and white container. The inspection shall be performed by a person with normal visual acuity, natural or corrected, under bright white light of no less than 100-foot candles. A general purpose two D-cell flashlight, with fresh batteries and lamp, positioned 18 inches directly above a surface will illuminate the surface with an intensity of no less than 100-foot candles.

4.5.11 Extended storage stability. Extended storage stability shall be determined by storing the cleaning composition for 1 year in accordance with ASTM F1104, except the packaging shall be that selected for delivering the cleaning composition to the Government. The cleaning composition shall then be tested as specified in table XII to determine compliance with 3.4.11 through 3.4.19.

4.5.12 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.4.3 as specified (see 6.2). The Navy and Marine Corps Force Health Protection Command (NMCFHPC) will evaluate the NOC using data provided by the manufacturer/distributor to the NMCFHPC (see 3.4.3 and 6.9).

4.5.13 Prohibited materials. Prohibited materials shall be verified as specified (see 3.4.8 and 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these

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personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The cleaning compound is intended for the precision cleaning of oxygen system piping and components, including valves, regulators, piping, flasks, and connectors. This cleaning compound is a critical item for cleaning oxygen systems and components during manufacture, maintenance, and repair. Common industry products use solvents, whereas this military-unique product utilizes an aqueous process. To be effective, this cleaning compound must be used in accordance with the oxygen cleaning process specified in MIL-STD-1330, or equal. The quality of this product directly affects the safe performance of oxygen systems. Current stock of NOC may be used until depleted; future requisitions for NOC should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article inspection is required (see 3.1 and 6.3).

c. Process for obtaining Technical Authority approval of alternative NOC manufacturing requirements (see 3.3.2).

d. Requirements for an HHA (see 3.4.3 and 4.5.12).

e. Requirement for prohibited materials verification (see 3.4.8 and 4.5.13).

f. Requirements for the submission of compatibility test procedures and compatibility tests for approval by the Technical Authority (see 3.4.8).

g. Government-approved aqueous oxygen cleaning process required for the service test, if other than the process specified in MIL-STD-1330 (see 3.4.21).

h. NOC lot formation, if other than as specified (see 4.4.1).

i. Packaging requirements (see 5.1 and 6.5).

6.3 First article inspection. Use of any cleaning composition that has not satisfactorily completed first article testing could result in catastrophic equipment failure and personnel injury or death. Letters of authorization for submittal of production samples for first article service tests may be obtained as specified (see 6.2). For cleaning compositions other than as specified in 3.3.1, the manufacturer should furnish copies of a first article inspection report. Included with the report should be:

a. Data showing that the material conforms to 3.2 (composition) and 3.4 (performance characteristics).

b. SDS prepared in accordance with FED-STD-313.

c. Certification that no change will be made to the product composition after the product has been shipped for first article service testing.

6.4 Conformance inspection. Failure to meet conformance requirements may result in the failure of the cleaning composition to perform as demonstrated during first article testing. This can result in the catastrophic equipment failure and personnel injury or death. Purchase of cleaning compositions covered by this specification should require manufacture in a certified facility in accordance with ISO 9001 or equivalent.

6.5 Packaging. Packaging that has not passed first article testing for extended storage stability (see 3.4.20) can contaminate the cleaning composition. This can cause hazardous flammable and toxic material to be introduced into the systems and components being cleaned resulting in catastrophic equipment failure and personnel injury or death. For the NOC composition cleaning compound, the only packaging that has passed first article testing for extended storage stability (see 3.4.20) was 5- and 55-gallon containers conforming to A-A-58020; where the interior steel surfaces are painted with a corrosion inhibiting primer; drums contain an unplasticized polyethylene liner which is not adhesively bonded to the drum interior, and prior to packaging, the interior of all containers were thoroughly spray cleaned with 120 to 150 °F (49 to 65 °C) demineralized water.

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6.6 Definitions.

6.6.1 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.6.2 Technical Authority approval. Test procedures are detailed step-by-step instructions for testing to be performed and include provisions for recording results. Test reports are test procedures that have the results recorded in the document along with any test witness signatures attesting that the test was conducted as written and that the results were accurately recorded; test witness signatures do not constitute approval or acceptance of the test results. Compatibility test procedures should be submitted to the Technical Authority (see 6.6.1) for review and approval prior to any testing to ensure that the testing will be correctly conducted in accordance with the provisions of this specification. Compatibility test reports should be submitted to the Technical Authority for review and acceptance of the test results. Those results will be considered for updating tables A-I through A-III of MIL-STD-1622 as part of future revisions to this specification.

6.7 Patent notice. The Government has a royalty-free license under the following listed patent for the benefit of manufacturers of the NOC composition cleaning compound either for the Government or for use in equipment to be delivered to the Government.

U.S. PATENT NUMBERS: 5,427,709 and 5,520,837

6.8 SDSs. Contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313. NOC composition cleaning compounds should include the following precautions, prepared by the Navy and Marine Corps Force Health Protection Command (NMCFHPC), in the SDS.

a. Skin contact with NOC will result in irritation and chemical burns. Wear impervious protective clothing constructed of neoprene or other material of equivalent resistance to penetration to prevent skin contact. Use elbow-length gloves (with cuffs) as minimum protection any time open containers of NOC are handled. Wash potentially exposed skin areas with soap and water at breaks and at the conclusion of operations. Do not use organic solvents for this purpose. Thoroughly clean protective garments at the conclusion of the operation and store for reuse. Remove clothing that becomes contaminated as soon as possible and thoroughly clean before reuse. Seek prompt medical assistance should a rash, chemical burn, or other adverse effect be experienced that may be related to working with NOC. Do not store or consume food and tobacco in areas where they could be contaminated with NOC.

b. Eye contact with NOC could result in chemical burns. Wear chemical worker's goggles for all operations where eye contact with NOC could occur. Use a full-length face shield for any operation where splashing of the material could occur. In the event of eye contact, personnel should flush their eyes thoroughly with freshwater for a minimum of 15 minutes, then seek prompt medical attention. Provide emergency eyewash systems near the work area, conforming to the design requirements of the American National Standards Institute (ANSI).

c. Over exposure to mists or vapors from NOC may result in sneezing, coughing, respiratory system irritation, and possibly chemical burns and subsequent edema in the upper airways. Based on off-gas testing at temperatures up to 180 °F (82 °C), such a possibility is unlikely considering the magnitude and frequency of use. In the event any adverse health effects are experienced, cease operations involving NOC and consult the cognizant medical department representative for additional guidance.

6.9 Toxicity evaluation. The NMCFHPC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCFHPC to the Government for evaluation. The HHA process is described on the NMCFHPC's website, https://www.med.navy.mil/Navy-and-Marine-Corps-Force-Health-Protection-Command/Environmental-Health/Industrial-Hygiene/Acquisition-Technical-Support/Health-Hazard-Assessments-HHA/.

6.10 National stock numbers (NSN) and sizes. The sizes and NSNs for NOC cleaning compositions are listed in table XIII.

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TABLE XIII. NOC sizes and NSNs.

Size/unit of issue	Military symbol/NATO code	NSN
5-gallon (CN)	None	6850-01-389-3859
55-gallon (DR)	None	6850-01-389-3880

6.11 Subject term (key word) listing.

Immersion
Off-gassing
Piping
Pumped cleaning
Spray
Ultrasonic cleaning

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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TEST PROCEDURE FOR NON-VOLATILE RESIDUE IN NAVY OXYGEN CLEANER BY GRAVIMETRIC ANALYSIS

A.1 SCOPE

A.1.1 Scope. This procedure covers the determination of non-volatile residue (NVR) in the Navy oxygen cleaner (NOC) by filtration and subsequent gravimetric analysis. The residue could consist of any insoluble low vapor pressure material in the NOC including particulate, gels, and inorganic as well as organic residues that are retained by a Teflon membrane filter with a pore diameter of 0.45 micrometer. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 APPLICABLE PUBLICATIONS

A.2.1 General. The publications listed in this section are specified in appendix A of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in appendix A of this specification, whether or not they are listed.

A.2.2 Government publications.

A.2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

A-A-59503 - Nitrogen, Technical

(Copies of this publication are available online at https://quicksearch.dla.mil.)

A.3 PROCEDURE

A.3.1 Summary of test method. The NVR in NOC is determined by passing a given quantity of NOC through a 0.45-micrometer Teflon filter and measuring the increase in the weight of the filter. The NOC sample is either pushed through the filter by pressurizing a filtration reservoir containing the NOC sample with oil-free nitrogen, or the NOC sample is drawn through the filter by a vacuum. The filter collects the NVR contained in the NOC. The filter is subsequently washed with demineralized water, dried, and weighed. The increase in the weight of the filter is a measure of the NVR in the NOC.

A.4 SIGNIFICANCE IN USE

A.4.1 NOC. NOC is an aqueous inorganic alkaline solution that is filtered to 0.1 micrometer during manufacture resulting in a material specification of 1.0 milligrams per liter (mg/L) for insoluble matter. NOC removes oil and grease by displacement with the resultant removed contaminants existing in NOC as a filterable suspension. This test method measures the amount of NVR contained in NOC. The test is used to verify the level of NVR in the NOC. The method will determine the amount of all types of NVR in NOC including organic material, particulate, and gels.

A.4.2 Filtrate drying. The filtrate is dried at a relatively low temperature. Therefore, this procedure is not expected to result in any appreciable loss of semi-volatile contaminates.

A.4.3 Verification. A variation of this test procedure is used to verify the cleanliness of Navy oxygen systems and components after cleaning with NOC.

A.4.4 Contamination. This analysis is susceptible to interference from demineralized water contaminated with insoluble material, from a filtration column contaminated with dried NOC residue or ambient dust, and to homogenizer wear that could contaminate the NOC sample with particulate.

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A.5 APPARATUS

A.5.1 Balance. Accurate to 0.1 mg with a zero drift less than 0.1 mg. A balance with an accuracy of 0.05 mg with a zero drift less than 0.05 mg is preferred. The balance shall be calibrated in accordance with the Navy Metrology and Calibration Program, or equivalent. Between calibrations, the accuracy of the balance should be verified using certified balance weights.

A.5.2 Balance weights. 1.0, 2.0, 5.0, and 10.0 mg, accurate to 0.02 mg (NIST Class S or ASTM Class 2) and made from non-magnetic material.

A.5.3 Membrane filter, pure Teflon, disposable. 0.45- or 0.50-micrometer pore size, 47-millimeter diameter. Bonded or laminated membrane filters shall not be used.

A.5.4 Membrane filter holder and filtration reservoir. Suitable to capture the 47-millimeter diameter filter specified in A.5.3, 250-mL capacity minimum, manufactured from chemically inert materials such as stainless steel or Teflon.

A.5.5 Laboratory homogenizer with generator. Homogenizer motor power of 500 to 1,000 watts with a no-load speed control of 10,000 to 30,000 revolutions per minute (rpm) and a 20-millimeter diameter saw tooth generator capable of homogenizing a 1-liter sample into a 1-5 micrometer emulsion at about 7,500 rpm loaded speed. A holder or stand for the homogenizer is recommended.

A.5.6 NOC sample/homogenizing bottles. 1-liter volume for collecting and homogenizing NOC samples. Ensure the opening is large enough to allow insertion of the homogenizer generator (item A.5.5). Sample bottles should be made from Teflon, Teflon-lined glass, Teflon-lined plastic, or safety coated (exterior) glass with Teflon-lined lid, or corrosion resistant metal.

A.5.7 Drying oven. Capable of maintaining a temperature of 131 to 140 °F (55 to 60 °C).

A.5.8 Graduated cylinder. 250-mL volume minimum, with minimum 50-mL graduations to measure 200 mL of homogenized NOC sample to be filtered, glass or polypropylene.

A.5.9 Vacuum collector. 500 mL to collect filtered NOC when using a vacuum to draw the NOC through the membrane filter; preferably plastic. The stopper used to seal the collector should be made of Teflon.

A.5.10 Beaker. 250 mL to collect the filtered NOC, preferably plastic, when pressurizing the NOC sample.

A.5.11 Beaker. 250 mL to measure 100 mL of demineralized water used to wash the membrane filter, plastic or glass.

A.5.12 Erlenmeyer flask. 1 liter for rinsing the homogenizer, plastic or glass. Ensure the opening is large enough to allow insertion of the homogenizer generator (item A.5.5).

A.5.13 Petri dish. Covered glass, for drying membrane filter in the oven.

A.5.14 Forceps. Non-serrated tips, to handle filters.

A.5.15 Eye dropper or pipette. To transfer isopropyl alcohol to wet the membrane filter prior to filtration.

A.5.16 Teflon tape. To prepare threaded connections on the filtration reservoir.

A.5.17 Vacuum source. Capable of drawing a vacuum of 10 inches Hg [5 psi differential (psid)] and controllable at 2-inch Hg (1 psid) increments; required if using a vacuum to draw the NOC through the membrane filter.

A.5.18 Nitrogen pressure source. Capable of providing 5 psig and controllable at 1-psi increments; required if using pressure to push the NOC through the membrane filter.

A.6 REAGANTS

A.6.1 Navy oxygen cleaner. In accordance with this specification.

A.6.2 Demineralized water. In accordance with ASTM D1193, type II.

A.6.3 Isopropyl alcohol. American Chemical Society (ACS) grade, having an NVR of 5 ppm (0.0005 percent) or less.

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A.6.4 Nitrogen. In accordance with A-A-59503, type I, grades A, B, or UHP, class 1, or equal.

A.7 SAMPLE

A.7.1 NOC sample. A 500- to 600-mL NOC sample shall be collected for NVR determination. The sample bottles used to collect the sample shall have been cleaned as specified in A.8. Care shall be exercised while collecting the sample to avoid contamination.

A.7.2 Homogenized NOC sample. A 200-mL sample of homogenized NOC will be used for the gravimetric analysis.

A.8 PREPARATION AND MAINTENCE OF APPARATUS

A.8.l Apparatus. Apparatus used in the procedure; including the sample bottles, homogenizer, homogenizing container, graduated cylinder, filtration reservoir, and demineralized water beaker; shall have been cleaned using a component cleaning process in accordance with MIL-STD-1330 or equivalent process.

A.8.2 Filter connections. Threaded connections in the filtration reservoir and filter support shall be wrapped with Teflon tape to prevent leakage of the NOC sample.

A.8.3 Rinsing. After contact with NOC, all apparatus shall immediately be rinsed with demineralized water to remove residual NOC, to prevent formation of inorganic deposits, and to maintain cleanliness.

A.9 PROCEDURE. The following procedure shall be performed:

a. Select a membrane filter that has equilibrated to ambient conditions.

b. Using forceps, weigh the filter to the nearest 0.1 mg, preferably to the nearest 0.05 mg.

c. Transfer the membrane filter to the filter holder and attach the filter holder to the filtration reservoir. Do not over tighten. Over tightening may cause permanent deformation, allowing the NOC sample to leak.

d. Homogenize the 500- to 600-mL NOC sample for 3 minutes at a speed of about 7,500 rpm.

e. Using an eyedropper or pipette, wet the filter with about 1 mL of isopropyl alcohol. Allow at least 15 seconds for the isopropyl alcohol to fully wet the filter. This is observable as a change in filter color from opaque white when dry to translucent when wet.

f. Optional – Excessive isopropyl alcohol can react with NOC to form a white precipitate that may cause high NVR results. If this interference is suspected, the isopropyl alcohol can be removed from the filter by transferring 10 to 20 mL of demineralized water to the filter reservoir. Filter the demineralized water by vacuum or by pressurizing the reservoir with oil-free nitrogen before proceeding to step A.9.g.

g. Transfer 200 mL of homogenized NOC to the filter reservoir.

h. Filter the homogenized NOC at less than 50 mL per minute by vacuum or by pressurizing the reservoir with oil-free nitrogen. Very little pressure differential (less than 5 psid) will be required to obtain the desired filtration rate. Do not exceed a filtration rate of 50 milliliters per minute.

i. Transfer 100 to 110 mL of demineralized water to the filter reservoir to wash the filter and reservoir of residual NOC.

j. Filter the demineralized water at less than 50 mL per minute by vacuum or by pressurizing the reservoir with oil-free nitrogen. Very little pressure differential (less than 5 psid) will be required to obtain the desired filtration rate. Do not exceed a filtration rate of 50 milliliters per minute.

k. Remove the filter, place in a petri dish, and dry with petri dish cover ajar in an oven at 131 to 140 °F (55 to 60 °C) for 20 to 30 minutes.

l. Remove the petri dish with filter from the oven, transfer the filter to an ambient temperature petri dish, and allow it to cool with cover ajar for at least 2 minutes.

m. Weigh the filter to the nearest 0.1 mg, preferably to the nearest 0.05 milligram.

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A.10 CALCULATIONS. The level of NVR shall be calculated as follows:

a. Subtract the initial weight of the filter (step A.9.b) from final weight of the filter (step A.9.m) after filtering the NOC. Record this weight difference in milligrams.

b. Calculate the level of NVR in the NOC by multiplying the net weight difference (step A.10.a) by five and report in mg/L.

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TEST PROCEDURE FOR NAVY OXYGEN CLEANER pH BY SPECTROPHOTOMETRIC ANALYSIS

B.1 SCOPE

B.1.1 Scope. This procedure covers the determination of Navy oxygen cleaner (NOC) pH by spectrophotometric analysis. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

B.2 PROCEDURE

B.2.1 Summary of test method. The pH of the NOC is determined by using a UV/Vis spectrometer to measure the absorbance of NOC with and without alizarin yellow R indicator solution at 491 nanometers (nm) and 385 nm. The difference in absorbance is computed at both wavelengths (absorbance of NOC + indicator - absorbance of NOC). The log of the ratio of the difference in absorbance at 491 nm and 385 nm is referenced to a calibration curve developed from primary and secondary pH standards to determine the pH of NOC.

B.3 SIGNIFICANCE AND USE

B.3.1 NOC. NOC is an aqueous inorganic alkaline cleaning solution having a pH of about 11.9. The solution is comprised of water, sodium silicate, sodium molybdate, and sodium fluoroborate. Accurate and consistent pH measurements of NOC have proven difficult to obtain when using hydronium ion concentration meters. The combination of sodium silicate, high pH, and high sodium ion concentration have resulted in erratic readings with both glass probes and ion selective field effect transistor (ISFET) probes.

B.3.2 NOC pH. The determination of the pH of NOC using a spectrophotometer is based on the methodology described in Robert-Baldo, Gillian, Morris, Michael J., and Byrne, Robert H., “Spectrophotometric Determination of Sea Water pH Using Phenol Red”, Analytical Chemistry, 1985, 57, 2564-2567. This method is based on a spectrophotometric calibration curve derived from pH standards in the range of 10.0 to 12.6. Alizarin yellow R, an indicator in the 10 to 12 pH range, was chosen for the NOC pH analysis. Alizarin yellow R exhibits two absorption maxima between 300 and 600 nm. At a pH of 10, the indicator has a maximum in the absorbance at approximately 375 nm. As the pH rises to 12, the absorption peak at 375 decreases and a peak near 491 nm arises. However, results obtained in the literature show that to obtain the best correlation between the change in pH from 10 to 12.5 and the change in the absorbance of the indicator, the log of the ratio of the absorbance at 491 nm and at 385 nm (not 375 nm) should be used.

B.3.3 Contamination. NOC mixed with tap water can introduce interfering ions such as calcium, which will invalidate the analysis results.

B.4 APPARATUS

B.4.1 UV/Vis spectrophotometer. Single or dual beam device with a wavelength range of at least 340 to 600 nm, a wavelength accuracy of at least ±2 nm and a precision of at least ±0.5 nm. The spectrophotometer shall be calibrated in accordance with the Navy Metrology and Calibration Program, or equivalent. Between calibrations, the accuracy of the spectrophotometer shall be verified with calibration filters.

B.4.2 Spectrophotometer cells. 10 mm path length absorption cells suitable for use in B.4.1. Matched cells shall be used with dual beam spectrophotometers. The cells should be of fused silica (quartz). Fused quartz has no absorption at 385 nm. Optical glass may be used. However, some absorption by the optical glass at 385 nm may be experienced.

B.4.3 pH meter. Accurate to at least plus or minus 0.05 pH units within a range of 10.0 to 12.6.

B.4.4 Balance. Accurate to 0.1 milligram with a zero drift less than 0.1 milligram.

B.4.5 Assorted lab ware. Rinse bottles, sample bottles, micro pipettes, and graduated cylinders.

B.5 REAGENTS

B.5.1 Navy oxygen cleaner. In accordance with this specification.

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B.5.2 Demineralized water. In accordance with ASTM D1193, type II.

B.5.3 Alizarin yellow R indicator solution. Saturated solution prepared by mixing 0.1 gram of alizarin yellow R to each 100 mL of demineralized water, allowing the mixture to stand until all undissolved solid has settled to the bottom of the container. The indicator solution is light-sensitive and shall be stored away from light in tightly sealed brown glass containers. The indicator solution has a shelf-life of 1 month.

B.5.4 0.2M potassium chloride. ACS reagent grade KC1 and demineralized water.

B.5.5 0.01M, 0.lM, and 0.2M sodium hydroxide. ACS reagent grade NaOH and demineralized water.

B.5.6 0.05M sodium phosphate, tribasic. ACS reagent grade Na_3PO_4 and demineralized water.

B.5.7 0.05M sodium phosphate, dibasic. ACS reagent grade Na_2HPO_4 and demineralized water.

B.5.8 0.025M sodium carbonate. ACS reagent grade Na_2CO_3 and demineralized water.

B.5.9 0.025M sodium bicarbonate. ACS reagent grade Na_2HCO_3 and demineralized water.

B.6 PROCEDURE FOR CALIBRATION CURVE. The following procedure shall be performed:

a. Prepare the pH standards listed in table B-I.

b. Measure and record the pH of each standard in accordance with ASTM E70.

c. For single beam spectrophotometers, perform the following:

(1) Measure and record the baseline absorbance of each pH standard at 385 nm and 491 nm.

(2) Add 300 microliters (0.3 mL) of alizarin yellow R indicator (see B.5.3) solution to each 25 mL of pH standard. With the same cell used in B.6.c(1) or a matched cell, measure and record the absorbance of each pH standard with indicator solution at 385 nm and 491 nm.

(3) Subtract the absorbance of the pH standards (see B.6.c(1)) from the absorbance of the pH standards with indicator solution (see B.6.c(2)) at 385 nm and 491 nm, respectively. Record these values as abs_{385} and abs_{491}.

d. For dual beam spectrophotometers, perform the following:

(1) Fill the reference beam cell with pH standard.

(2) Fill the sample beam cell with pH standard containing indicator solution prepared by adding 300 microliters (0.3 mL) of alizarin yellow R indicator (see B.5.3) solution to each 25 mL of pH standard.

(3) The absorbance of each pH standard is measured at 385 nm and 491 nm. Record these values as abs_{385} and abs_{491}.

e. Tabulate the data so that the pH measurements from B.6.b is the Y axis, and the corresponding data from B.6.c(3) or B.6.d(3), represented as the $[\log(abs_{491}/abs_{385})]$, is the X axis.

f. Perform a linear regression of the data from B.6.e, and determine the equation of the line that best fits the data. A carefully prepared calibration curve should result in a correlation coefficient of 0.98±0.02.

g. Record an equation for pH as a function of the log of the ratio of the two absorbances that yield:

$$pH = b + m[\log(abs_{491}/abs_{385})]$$

where: b is a constant representing the Y axis cross-connect determined from B.6.f
m is a constant representing the slope of the line determined from B.6.f

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TABLE B-I. pH standards.

Reported pH 1/	Reagents	Composition (volume %)
12.6	0.20M KC1	25.0
	0.20M NaOH	25.6
	Water	49.4
12.6	0.05M NaOH	100.0
11.9	0.01M NaOH	100.0
11.7	0.05M Na_3PO_4	100.0
11.5	0.05M Na_2HPO_4	50.0
	0.10M NaOH	11.1
	Water	38.9
11.3	0.05M Na_2HPO_4	50.0
	0.10M NaOH	7.6
	Water	42.4
10.9	0.05M Na_2HPO_4	50.0
	0.10M NaOH	3.3
	Water	46.7
10.0	0.025M Na_2CO_3	50.0
	0.025M $NaHCO_3$	50.0

NOTE:

1/ The 12.6 KC1/NaOH, 11.5, 11.3, and 10.9 pH standards were prepared as described in Bates, Roger G. and Bower, Vincent E., “Alkaline Solutions for pH Control”, Analytical Chemistry, 1956, 28, 1322-1324.

B.7 PROCEDURE FOR pH OF NOC. The following procedure shall be performed:

a. Obtain a 25-mL, or greater, sample of NOC at room temperature (73±5 °F [23±3 °C]).

b. For single beam spectrophotometers, perform the following:

(1) Measure and record the baseline absorbance of NOC at 385 nm and 491 nm.

(2) Add 300 microliters (0.3 mL) of alizarin yellow R indicator (see B.5.3) solution to each 25 mL of NOC. With the same cell used in B.7.b(1) or a matched cell, measure and record the absorbance of NOC with indicator solution at 385 nm and 491 nm.

(3) Subtract the absorbance of NOC without indicator (see B.7.b(1)) from the absorbance of NOC with indicator solution (see B.7.b(2)) at 385 nm and 491 nm, respectively. Record these values as abs_{385} and abs_{491}.

c. For dual beam spectrophotometers, perform the following:

(1) Fill the reference beam cell with NOC.

(2) Fill the sample beam cell with NOC containing indicator solution prepared by adding 300 microliters (0.3 mL) of alizarin yellow R indicator (see B.5.3) solution to each 25 mL of NOC.

(3) The absorbance of each pH standard is measured at 385 nm and 491 nm. Record these values as abs_{385} and abs_{491}.

d. Determine pH using the equation from B.6.g and the values for abs_{385} and abs_{491} from B.8.b(3) or B.8.c(3). The accuracy of this analysis is ±0.1 pH units.

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TEST PROCEDURE FOR NAVY OXYGEN CLEANER TOTAL ALKALINITY BY TITRATION

C.1 SCOPE

C.1.1 Scope. This procedure covers the determination of the total alkalinity (quantity of base) contained in Navy oxygen cleaner (NOC) referenced to a pH of about 10.0. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

C.2 APPLICABLE PUBLICATIONS

C.2.1 General. The publications listed in this section are specified in appendix C of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in appendix C of this specification, whether or not they are listed.

C.2.2 Government publications.

C.2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-I-735 - Isopropyl Alcohol

(Copies of this publication are available online at https://quicksearch.dla.mil.)

C.2.3 Non-Government publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E898 - Standard Practice for Calibration of Non-Automatic Weighing Instruments

(Copies of this publication are available online at www.astm.org.)

C.3 PROCEDURE

C.3.1 Summary of test method. The total alkalinity of the NOC is determined by measuring the amount of NOC that is necessary to neutralize a standard acid solution of potassium acid phthalate (KHP) in pure water. Thymolphthalein is used as the end-point indicator (pH 10.0).

C.4 SIGNIFICANCE AND USE. As addressed in appendix B of this specification, measurement of NOC pH with a hydronium ion concentration meter is extremely difficult. This test method measures the total alkalinity contained in NOC without the use of a pH meter. As the alkalinity of the NOC is depleted, the pH of the cleaner will decrease, and at a pH of 11.0, the cleaner will have lost most of the properties necessary for it to be an effective cleaner. Determining the total alkalinity at the time of manufacture will provide a measure of the cleaner's stability and capacity to be recycled and reused in the field. Determining the total alkalinity in the field can be a measure of the remaining useful life of the cleaner.

C.5 APPARATUS

C.5.1 Balance. Accurate to 0.001 gram with a zero drift less than 0.001 gram and a capacity sufficient for the preparation of the KHP and thymolphthalein pH indicator solutions as specified in C.7. The accuracy of the balance should be verified using ASTM E3898 or an equivalent method.

C.5.2 Erlenmeyer flask. Plastic or glass, 100 to 250 mL, for dissolving the KHP and performing the titration. Suitable beakers or volumetric containers may be used in place of the Erlenmeyer flask.

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C.5.3 Graduated cylinder. Plastic or glass, 50 to 100 mL, for measuring the volume of demineralized water used to dissolve the KHP, and for measuring the volume of isopropyl alcohol for making the indicator solution.

C.5.4 Eyedropper or pipette. Glass or plastic, for adding pH indicator to the KHP solution.

C.5.5 Buret. Capacity 50 mL, with 0.1-mL graduation intervals, plastic or glass, for measuring the volume of NOC needed to neutralize the KHP solution.

C.6 REAGENTS

C.6.1 Navy oxygen cleaner. In accordance with this specification.

C.6.2 Demineralized water. In accordance with ASTM D1193, type II.

C.6.3 Potassium acid phthalate or potassium biphthalate (KHP). Primary standard or acidimetric standard, crystals, ACS reagent grade or better, CAS No. 877-24-7. Do not use powdered KHP since it does not completely dissolve.

C.6.4 Thymolphthalein. pH indicator, powder, CAS No. 125-20-2.

C.6.5 Isopropyl alcohol. In accordance with TT-I-735 or equal.

C.7 REAGENT PREPARATION. The following preparation shall be performed:

C.7.1 KHP solution. Prepare a 4.0 percent wt/vol KHP solution by adding 100.0±1.0 mL of demineralized water (see C.6.2) to each 4.0±0.1 grams of dry KHP (see C.6.3). The KHP may not dissolve in the water immediately. After adding the water, gently swirl or stir the contents for about 1 minute being careful not to deposit solid KHP on the sides of the flask. Let the flask stand for about 10 minutes and swirl or stir the contents for an additional minute. The KHP solution has a 3-month shelf life when stored in a tightly sealed glass container.

C.7.2 pH indicator solution. Prepare a 0.2 percent wt/vol thymolphthalein pH indicator solution by adding 50±1.0 mL of isopropyl alcohol (see C.6.5) to each 0.1±0.01 gram of thymolphthalein (see C.6.4). The solution may have to sit for about 4 hours to allow the thymolphthalein to completely dissolve. The pH indicator solution should be clear, water white with no precipitate. The pH indicator solution has a 1-month shelf life when stored in a tightly sealed brown glass or plastic container.

C.8 PROCEDURE. The following procedure shall be performed:

a. Transfer 50.0±1.0 mL of the 4.0 percent KHP solution (see C.7.1) into the Erlenmeyer flask.

b. Add five drops of the 0.2 percent thymolphthalein pH indicator solution (see C.7.2) into the Erlenmeyer flask. The KHP solution should be clear.

c. Fill a 50-mL buret with NOC test solution. Record to the nearest 0.1 mL the level of the solution in the buret.

d. Slowly add NOC to the KHP solution while swirling the flask until a blue color endpoint persists for 30 seconds. Record to the nearest 0.1 mL the final level of NOC solution in the buret.

C.9 CALCULATION. The final level (from C.8.d) of NOC solution in the buret shall be subtracted from the initial level (from C.8.d) of NOC solution in the buret. This volume difference shall be recorded in milliliters. This number shall be reported as the total alkalinity - mL of NOC.

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CONCLUDING MATERIAL

Custodians:
Army – EA
Navy – SH
Air Force – 68

Preparing activity:
Navy – SH
(Project 6850-2026-002)

Review activities:
Army – MI
Navy – AS
DLA – GS

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